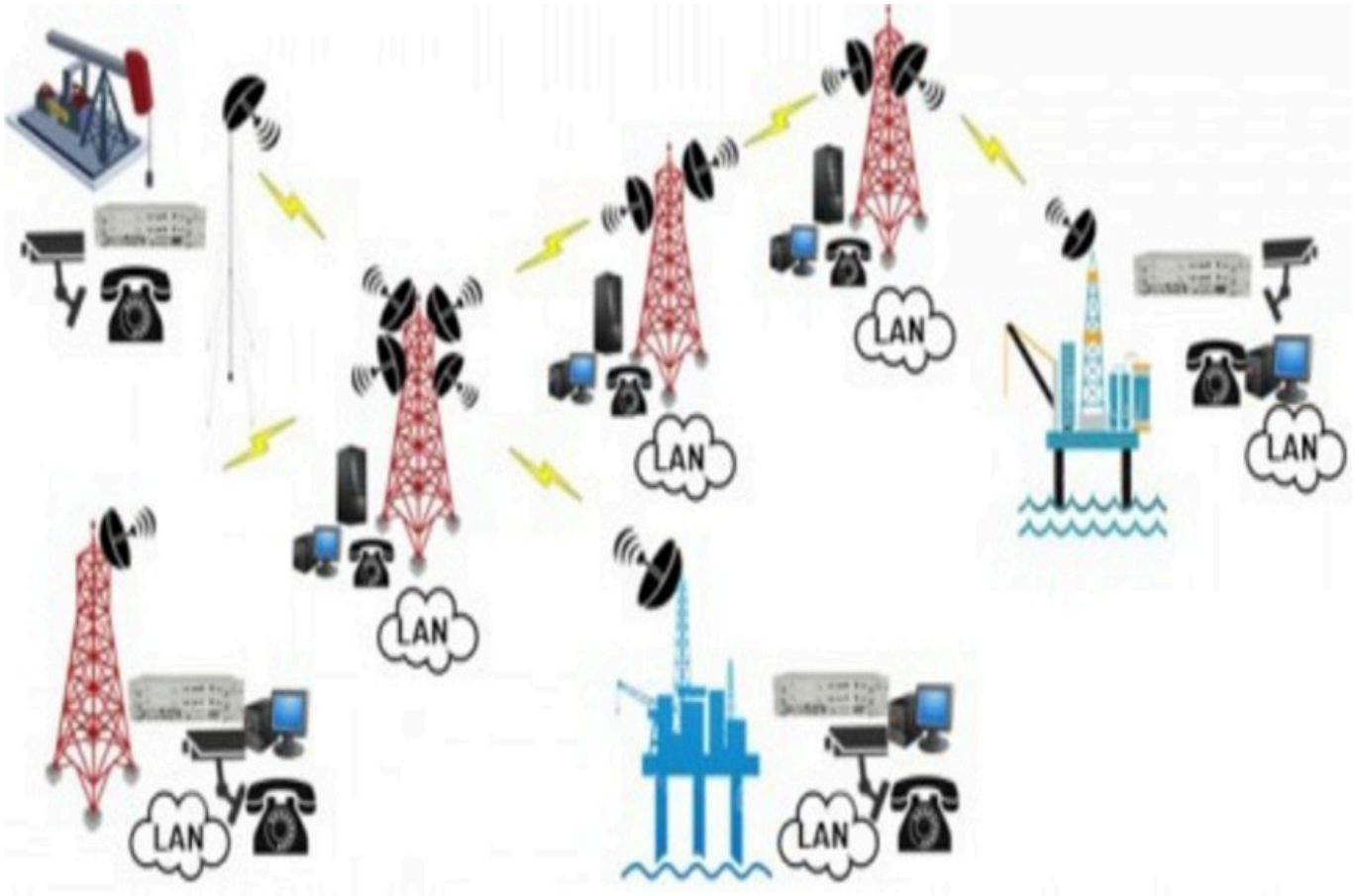


Qualification Pack



Optical Network Terminal Technician

QP Code: TEL/Q6207

Version: 5.0

NSQF Level: 4

Telecom || 3rd Floor, Plot No 126, Sector - 44
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Qualification Pack

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TEL/Q6207: Optical Network Terminal Technician

Brief Job Description

An Optical Network Terminal Technician is responsible for ensuring the continuous 24x7 operation of Optical Network Terminal (ONT) sites. The role involves performing routine maintenance, troubleshooting and repairing basic faults at ONT sites, promoting digital adoption among the local population, and delivering telecom services efficiently.

Personal Attributes

The individual should be self-disciplined, proactive, and a strong team player with a problem-solving mindset. The person should possess analytical skills, effective communication abilities, and the ability to work efficiently under pressure.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [TEL/N6226: Maintain Site Security and Hygiene](#)
2. [TEL/N6227: Perform Preventive Maintenance of Optical Network Terminal \(ONT\) Components](#)
3. [TEL/N6228: Promote use of Devices and Provide Services](#)
4. [TEL/N9109: Follow sustainable practices in telecom infrastructure management](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Telecom
Sub-Sector	Network Managed Services
Occupation	Network Operation and Maintenance
Country	India
NSQF Level	4
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7422.0202

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Minimum Educational Qualification & Experience	12th grade Pass OR Completed 2nd year of the 3-year diploma after 10 (in Electronics and Communication Engineering/ Telecommunication Engineering/ Networking Technology) OR 10th grade pass with 3 Years of experience in the operations and maintenance of passive infrastructure OR Previous relevant Qualification of NSQF Level (3.0) with 3 Years of experience in the operations and maintenance of passive infrastructure OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience in the operations and maintenance of passive infrastructure
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	17 Years
Last Reviewed On	NA
Next Review Date	30/04/2028
NSQC Approval Date	08/05/2025
Version	5.0
Reference code on NQR	QG-04-TL-04080-2025-V2-TSSC
NQR Version	2

Qualification Pack

TEL/N6226: Maintain Site Security and Hygiene

Description

This unit covers maintaining the security, hygiene, and sustainability of an Optical Network Terminal (ONT) site, including renewable energy integration and eco-friendly deployment practices

Scope

The scope covers the following :

- Coordinate activities for site security, hygiene, and environmental compliance
- Monitor equipment and ensure renewable energy integration

Elements and Performance Criteria

Coordinate activities for site security, hygiene, and environmental compliance

To be competent, the user/individual on the job must be able to:

- PC1.** perform custodial functions for an Optical Network Terminal (ONT) site, ensuring security, hygiene, and proper functioning
- PC2.** communicate with Optical Line Terminal (OLT) and Network Operation Center (NOC) supervisors to report issues or request technological upgrades
- PC3.** coordinate with Gram Panchayat officials and other local authorities for the smooth operation and security of the ONT site
- PC4.** follow eco-friendly site management practices, to ensure adherence to environmental compliance standards

Monitor equipment and ensure renewable energy integration

To be competent, the user/individual on the job must be able to:

- PC5.** check the functionality of Charge Controller Unit (CCU), Solar Photovoltaic (SPV) system, Telephone Junction Box (TJB), battery bank, and other components
- PC6.** maintain hygiene and operational standards for ONT site equipment, including renewable energy solutions
- PC7.** check all cable connectors for secure connections and detect any signs of damage
- PC8.** perform necessary steps for the repair or replacement of damaged cables, ensuring compliance with safety protocols
- PC9.** monitor solar-powered energy sources and battery banks to optimize performance and ensure uninterrupted power supply
- PC10.** report faults that cannot be rectified to the supervisor for further action
- PC11.** use fire extinguishers and follow emergency protocols in case of fire or other hazards
- PC12.** record faults, maintenance activities, and energy efficiency parameters in the prescribed format

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** standard guidelines to ensure uninterrupted operation and security of an Optical Network Terminal (ONT) site
- KU2.** escalation procedures for reporting issues with equipment or renewable energy systems
- KU3.** functionality and maintenance of Charge Controller Unit (CCU), Solar Photovoltaic System (SPV), Telephone Junction Box (TJB), battery bank, and fire extinguishers
- KU4.** best practices for ONT site hygiene, security, and environmental sustainability
- KU5.** basics of electrical wiring, renewable energy integration, and power backup management
- KU6.** steps to be followed for setting up, maintaining, and troubleshooting renewable energy solutions for telecom infrastructure
- KU7.** environmental impact reduction techniques during network setup, including responsible e-waste recycling and disposal
- KU8.** types, functions, and safe usage of fire extinguishers for different emergency situations
- KU9.** industry standards and regulatory compliance requirements for maintaining telecom site security, hygiene, and environmental sustainability
- KU10.** reporting and documentation procedures for equipment maintenance, repairs, and environmental compliance

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret equipment manuals, safety guidelines, and environmental regulations
- GS2.** prioritize tasks to ensure efficient operation and sustainability of the ONT site
- GS3.** maintain effective communication with OLT/NOC supervisors and local authorities
- GS4.** work systematically with attention to detail while handling security, hygiene, and renewable energy components
- GS5.** use digital tools and software to document maintenance activities, energy consumption, and environmental compliance
- GS6.** follow emergency response protocols, including fire safety measures and power failure management
- GS7.** implement safe work practices while handling electrical systems, solar equipment, and hazardous materials
- GS8.** apply problem-solving skills to diagnose and address equipment faults, cable damage, or renewable energy inefficiencies
- GS9.** train and guide site personnel on basic security, hygiene, and sustainable energy practices
- GS10.** maintain accurate records of equipment maintenance, energy performance, and e-waste disposal as per standard procedures

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Coordinate activities for site security, hygiene, and environmental compliance</i>	14	24	-	8
PC1. perform custodial functions for an Optical Network Terminal (ONT) site, ensuring security, hygiene, and proper functioning	3	5	-	2
PC2. communicate with Optical Line Terminal (OLT) and Network Operation Center (NOC) supervisors to report issues or request technological upgrades	3	6	-	2
PC3. coordinate with Gram Panchayat officials and other local authorities for the smooth operation and security of the ONT site	4	6	-	2
PC4. follow eco-friendly site management practices, to ensure adherence to environmental compliance standards	4	7	-	2
<i>Monitor equipment and ensure renewable energy integration</i>	16	26	-	12
PC5. check the functionality of Charge Controller Unit (CCU), Solar Photovoltaic (SPV) system, Telephone Junction Box (TJB), battery bank, and other components	2	3	-	1
PC6. maintain hygiene and operational standards for ONT site equipment, including renewable energy solutions	2	4	-	1
PC7. check all cable connectors for secure connections and detect any signs of damage	2	2	-	2
PC8. perform necessary steps for the repair or replacement of damaged cables, ensuring compliance with safety protocols	2	3	-	1
PC9. monitor solar-powered energy sources and battery banks to optimize performance and ensure uninterrupted power supply	2	4	-	2
PC10. report faults that cannot be rectified to the supervisor for further action	2	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. use fire extinguishers and follow emergency protocols in case of fire or other hazards	2	4	-	2
PC12. record faults, maintenance activities, and energy efficiency parameters in the prescribed format	2	3	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N6226
NOS Name	Maintain Site Security and Hygiene
Sector	Telecom
Sub-Sector	Network Managed Services
Occupation	Network Operation and Maintenance
NSQF Level	4
Credits	5
Version	5.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

TEL/N6227: Perform Preventive Maintenance of Optical Network Terminal (ONT) Components

Description

This unit covers the execution of preventive maintenance activities and real-time monitoring at ONT sites to ensure 24/7 operational efficiency and fault resolution

Scope

The scope covers the following :

- Perform preventive maintenance and maintain records
- Monitor equipment, use advanced testing tools, and report issues

Elements and Performance Criteria

Perform preventive maintenance and keep record

To be competent, the user/individual on the job must be able to:

- PC1.** read and follow preventive maintenance guidelines for Charge Controller Unit (CCU), Solar Photo Voltaic (SPV) System, Telephone Junction Box (TJB), battery bank, and fire extinguisher
- PC2.** conduct scheduled preventive maintenance of CCU, SPV, TJB, and battery bank using the prescribed checklists
- PC3.** ensure the availability of required equipment, including spectrum analyzers, OTDRs, and network testers
- PC4.** perform routine diagnostic checks using real-time remote monitoring tools to detect faults in ONT components
- PC5.** document maintenance activities in the preventive maintenance report form in the specified format

Monitor equipment, use advanced testing tools, and report issues

To be competent, the user/individual on the job must be able to:

- PC6.** observe the CCU panel, interpret alarms, and take corrective actions as per standard operating procedures
- PC7.** inspect battery bank cells and SPV panels for premature ageing and replace defective units as per manufacturer guidelines
- PC8.** examine and test patch cords, pigtails, and electrical wiring for damage and replace faulty components
- PC9.** conduct interference analysis using spectrum analyzers and OTDRs to identify signal loss and network degradation issues
- PC10.** report major equipment faults and alarm triggers to the OLT/NOC supervisor through the designated escalation matrix

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** standard guidelines for preventive maintenance of ONT components
- KU2.** criticality of ensuring continuous ONT operation and network uptime
- KU3.** the importance of adhering to scheduled maintenance tasks and documentation requirements
- KU4.** troubleshooting procedures for ONT faults, including remote diagnosis and real-time monitoring tools
- KU5.** organizational protocols for incident reporting, documentation, and escalation
- KU6.** the components, specifications, and functionality of Charge Controller Unit (CCU), Solar Photo Voltaic (SPV) System, Telephone Junction Box (TJB), battery bank, and fire extinguisher
- KU7.** types, specifications, and installation standards of electrical wiring, patch cords, and pigtails used in ONT systems
- KU8.** procedures for using spectrum analyzers, Optical Time-Domain Reflectometers (OTDRs), and network testers for fault detection and resolution
- KU9.** common interference issues in optical networks and methods to diagnose and rectify them
- KU10.** industry best practices for network maintenance, safety standards, and compliance with regulatory guidelines

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret preventive maintenance guidelines, equipment manuals, and standard operating procedures
- GS2.** maintain detailed records of maintenance activities and diagnostic results
- GS3.** communicate clearly with the supervisor and team members regarding maintenance tasks and equipment issues
- GS4.** work systematically with attention to detail to prevent equipment failures and network disruptions
- GS5.** collaborate with cross-functional teams, including NOC engineers and field technicians, to resolve faults efficiently
- GS6.** adapt to the use of advanced testing and diagnostic tools for fault identification and performance analysis
- GS7.** respond effectively to network alarms and take appropriate corrective actions within stipulated timelines
- GS8.** follow established safety procedures while handling ONT equipment and electrical components
- GS9.** develop problem-solving skills to address technical challenges in ONT maintenance
- GS10.** stay updated with new technologies and industry advancements related to fiber-optic network maintenance

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform preventive maintenance and keep record</i>	18	23	-	10
PC1. read and follow preventive maintenance guidelines for Charge Controller Unit (CCU), Solar Photo Voltaic (SPV) System, Telephone Junction Box (TJB), battery bank, and fire extinguisher	3	5	-	2
PC2. conduct scheduled preventive maintenance of CCU, SPV, TJB, and battery bank using the prescribed checklists	5	6	-	2
PC3. ensure the availability of required equipment, including spectrum analyzers, OTDRs, and network testers	4	4	-	2
PC4. perform routine diagnostic checks using real-time remote monitoring tools to detect faults in ONT components	3	4	-	2
PC5. document maintenance activities in the preventive maintenance report form in the specified format	3	4	-	2
<i>Monitor equipment, use advanced testing tools, and report issues</i>	12	27	-	10
PC6. observe the CCU panel, interpret alarms, and take corrective actions as per standard operating procedures	2	4	-	2
PC7. inspect battery bank cells and SPV panels for premature ageing and replace defective units as per manufacturer guidelines	2	5	-	2
PC8. examine and test patch cords, pigtails, and electrical wiring for damage and replace faulty components	3	7	-	2
PC9. conduct interference analysis using spectrum analyzers and OTDRs to identify signal loss and network degradation issues	3	5	-	2
PC10. report major equipment faults and alarm triggers to the OLT/NOC supervisor through the designated escalation matrix	2	6	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N6227
NOS Name	Perform Preventive Maintenance of Optical Network Terminal (ONT) Components
Sector	Telecom
Sub-Sector	Network Managed Services
Occupation	Network Operation and Maintenance
NSQF Level	4
Credits	6
Version	5.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

TEL/N6228: Promote use of Devices and Provide Services

Description

This unit covers enabling the local population to adopt and efficiently use digital devices (smartphones, tablets, e-terminals) by providing hands-on support, configuring devices, troubleshooting technical issues, and managing billing and revenue collection

Scope

The scope covers the following :

- Facilitate Adoption of Digital Devices
- Install, configure, and troubleshoot devices
- Manage revenue collection and site payments

Elements and Performance Criteria

Facilitate adoption of digital devices

To be competent, the user/individual on the job must be able to:

- PC1.** identify and engage target groups to introduce digital devices and their benefits
- PC2.** promote and recommend relevant devices based on user requirements and digital literacy levels
- PC3.** conduct on-site demonstrations of device functionalities to enhance user confidence and adoption
- PC4.** address concerns and misconceptions related to device usability, security, and affordability

Install, configure, and troubleshoot devices

To be competent, the user/individual on the job must be able to:

- PC5.** set up and install necessary equipment such as routers, modems, UPS, and cables at designated locations
- PC6.** configure smartphones, tablets, and e-terminals as per service provider guidelines
- PC7.** assist users in setting up accounts, applications, and network connectivity on digital devices
- PC8.** perform basic troubleshooting for common issues related to hardware, software, and network connectivity
- PC9.** guide users on security measures, software updates, and device maintenance to ensure long-term usability

Manage revenue collection and site payments

To be competent, the user/individual on the job must be able to:

- PC10.** distribute bills for service usage as per billing cycles
- PC11.** collect payments and issue receipts and maintain transaction records
- PC12.** maintain and update rent agreements and electricity bill records for ONT sites
- PC13.** coordinate rent, electricity, or service fees payments with the supervisor to avoid disruptions

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** the importance of digital literacy and its impact on social and economic inclusion
- KU2.** the types, specifications, and capabilities of smartphones, tablets, and e-terminals
- KU3.** role and benefits of broadband, internet services, and e-governance initiatives
- KU4.** installation and operational requirements of modems, routers, UPS, and networking cables
- KU5.** the steps for configuring operating systems, mobile applications, and security settings on digital devices
- KU6.** troubleshooting techniques for network, software, and hardware issues in digital devices
- KU7.** billing processes, payment collection methods, and compliance requirements
- KU8.** documentation standards for recording payments, agreements, and service usage
- KU9.** customer service principles, including conflict resolution and complaint handling
- KU10.** data protection measures and ethical handling of user information

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively in the local language and adapt messaging based on user demographics
- GS2.** engage and persuade users by explaining the practical benefits of digital services
- GS3.** document installation, configuration, and troubleshooting activities as per the required format
- GS4.** maintain a professional and supportive approach while resolving user concerns
- GS5.** coordinate with service providers, vendors, and supervisors for technical support and escalation
- GS6.** follow data security best practices to safeguard user information and transactions
- GS7.** adapt to evolving technologies and digital services to provide up-to-date assistance
- GS8.** manage time efficiently to handle multiple service requests and revenue collection tasks
- GS9.** use digital tools such as billing software, spreadsheets, and mobile applications for tracking records
- GS10.** maintain long-term user engagement by offering continued support and troubleshooting guidance

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Facilitate adoption of digital devices</i>	10	17	-	6
PC1. identify and engage target groups to introduce digital devices and their benefits	3	4	-	1
PC2. promote and recommend relevant devices based on user requirements and digital literacy levels	2	4	-	2
PC3. conduct on-site demonstrations of device functionalities to enhance user confidence and adoption	2	5	-	1
PC4. address concerns and misconceptions related to device usability, security, and affordability	3	4	-	2
<i>Install, configure, and troubleshoot devices</i>	10	19	-	9
PC5. set up and install necessary equipment such as routers, modems, UPS, and cables at designated locations	2	3	-	2
PC6. configure smartphones, tablets, and e-terminals as per service provider guidelines	2	4	-	2
PC7. assist users in setting up accounts, applications, and network connectivity on digital devices	2	4	-	2
PC8. perform basic troubleshooting for common issues related to hardware, software, and network connectivity	2	4	-	1
PC9. guide users on security measures, software updates, and device maintenance to ensure long-term usability	2	4	-	2
<i>Manage revenue collection and site payments</i>	10	14	-	5
PC10. distribute bills for service usage as per billing cycles	2	3	-	1
PC11. collect payments and issue receipts and maintain transaction records	3	4	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. maintain and update rent agreements and electricity bill records for ONT sites	2	3	-	2
PC13. coordinate rent, electricity, or service fees payments with the supervisor to avoid disruptions	3	4	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	TEL/N6228
NOS Name	Promote use of Devices and Provide Services
Sector	Telecom
Sub-Sector	Network Managed Services
Occupation	Network Operation and Maintenance
NSQF Level	4
Credits	4
Version	5.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

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TEL/N9109: Follow sustainable practices in telecom infrastructure management

Description

This unit focuses on sustainability practices in telecom infrastructure management, covering e-waste management, green practices, hazardous and non-hazardous waste, and energy efficiency.

Scope

The scope covers the following :

- Manage hazardous and e-waste at telecom sites
- Adopt green energy and resource efficiency practices
- Follow waste reduction strategies
- Ensure compliance with environmental regulations

Elements and Performance Criteria

Manage hazardous and e-waste at telecom sites

To be competent, the user/individual on the job must be able to:

- PC1.** identify, segregate, and categorize e-waste (batteries, cables, outdated equipment) and hazardous waste (lead-acid batteries, diesel residues, etc.)
- PC2.** dispose of or recycle waste following the applicable e-waste management guidelines
- PC3.** follow safe handling procedures for hazardous materials, including protective gear usage
- PC4.** maintain logs and records of disposed, recycled, or repurposed telecom waste

Adopt green energy and resource efficiency practices

To be competent, the user/individual on the job must be able to:

- PC5.** optimize power usage through energy-efficient telecom equipment (e.g., LED lighting, smart cooling systems)
- PC6.** assist in adopting solar-powered telecom towers and integration of hybrid energy systems (e.g. solar power and battery backup)
- PC7.** monitor and minimize fuel consumption in Diesel Generators (DG) sets through load balancing and regular maintenance

Follow waste reduction strategies

To be competent, the user/individual on the job must be able to:

- PC8.** reduce packaging waste by promoting the reuse of telecom materials and accessories
- PC9.** follow rainwater harvesting and wastewater reuse practices in telecom cooling systems
- PC10.** promote structured cabling to reduce cable waste and optimize resource utilization

Ensure compliance with environmental regulations

To be competent, the user/individual on the job must be able to:

- PC11.** follow Central Pollution Control Board (CPCB) guidelines for telecom site waste disposal
- PC12.** assist in conducting periodic environmental audits to ensure sustainability compliance
- PC13.** guide co-workers on eco-friendly practices and waste management policies at telecom sites

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** e-waste management rules (2022) applicable the telecom sector
- KU2.** Central Pollution Control Board (CPCB) hazardous waste disposal regulations
- KU3.** safety standards for battery handling and disposal (lead-acid, lithium-ion)
- KU4.** the importance of structured cabling and cable reuse to minimize telecom waste
- KU5.** the techniques for energy optimization in telecom infrastructure (e.g., smart cooling, LED lighting, hybrid power systems)
- KU6.** the role of solar energy and other renewable sources in reducing telecom carbon footprint
- KU7.** green telecom practices like fuel efficiency in DG sets and power-saving measures
- KU8.** best practices in telecom tower site waste management
- KU9.** the principles of water conservation and sustainable telecom site design
- KU10.** the importance of training telecom employees on environmental awareness and compliance

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** explain sustainability practices to team members
- GS2.** identify and address issues in waste management and energy efficiency
- GS3.** maintain logs on waste disposal, recycling, and energy savings
- GS4.** analyze power consumption and propose efficiency improvements
- GS5.** select between disposal, recycling, or repurposing of telecom waste
- GS6.** adhere to the applicable environmental laws and guidelines
- GS7.** work with vendors and waste management partners for sustainable practices
- GS8.** adapt to new green technologies and sustainability policies
- GS9.** ensure prompt and efficient waste disposal and as per applicable schedules

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Manage hazardous and e-waste at telecom sites</i>	8	15	-	5
PC1. identify, segregate, and categorize e-waste (batteries, cables, outdated equipment) and hazardous waste (lead-acid batteries, diesel residues, etc.)	2	4	-	2
PC2. dispose of or recycle waste following the applicable e-waste management guidelines	2	4	-	1
PC3. follow safe handling procedures for hazardous materials, including protective gear usage	2	4	-	1
PC4. maintain logs and records of disposed, recycled, or repurposed telecom waste	2	3	-	1
<i>Adopt green energy and resource efficiency practices</i>	6	15	-	5
PC5. optimize power usage through energy-efficient telecom equipment (e.g., LED lighting, smart cooling systems)	2	5	-	2
PC6. assist in adopting solar-powered telecom towers and integration of hybrid energy systems (e.g. solar power and battery backup)	2	5	-	1
PC7. monitor and minimize fuel consumption in Diesel Generators (DG) sets through load balancing and regular maintenance	2	5	-	2
<i>Follow waste reduction strategies</i>	8	10	-	5
PC8. reduce packaging waste by promoting the reuse of telecom materials and accessories	3	4	-	2
PC9. follow rainwater harvesting and wastewater reuse practices in telecom cooling systems	3	3	-	2
PC10. promote structured cabling to reduce cable waste and optimize resource utilization	2	3	-	1
<i>Ensure compliance with environmental regulations</i>	8	10	-	5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. follow Central Pollution Control Board (CPCB) guidelines for telecom site waste disposal	3	4	-	2
PC12. assist in conducting periodic environmental audits to ensure sustainability compliance	3	3	-	2
PC13. guide co-workers on eco-friendly practices and waste management policies at telecom sites	2	3	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	TEL/N9109
NOS Name	Follow sustainable practices in telecom infrastructure management
Sector	Telecom
Sub-Sector	
Occupation	Generic
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TEL/N6226.Maintain Site Security and Hygiene	30	50	-	20	100	25
TEL/N6227.Perform Preventive Maintenance of Optical Network Terminal (ONT) Components	30	50	-	20	100	25
TEL/N6228.Promote use of Devices and Provide Services	30	50	-	20	100	25
TEL/N9109.Follow sustainable practices in telecom infrastructure management	30	50	0	20	100	15
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	140	230	0	80	450	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NCVET	National Council for Vocational Education and Training
QP	Qualification Pack
MC	Model Curriculum
NSQF	National Skills Qualification Framework
NSQC	National Skills Qualification Committee
NOS	National Occupational Standards
NCO	National Classification of Occupations
ES	Employability Skills
TSSC	Telecom Sector Skill Council
ONT	Optical Network Terminal
OLT	Optical Line Terminal
NOC	Network Operation Center
CCU	Charge Controller Unit
SPV	Solar Photovoltaic
TJB	Telephone Junction Box
OTDR	Optical Time Domain Reflectometer
UPS	Uninterruptible Power Supply
DG	Diesel Generators
CPCB	Central Pollution Control Board

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.